Preferences for Product Variety in Targeted Pricing

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1 Introduction

Much of daily life takes place online. Shopping, socializing, and entertainment have become increasingly digital experiences. As such, firms have evolved to best optimize according to their objectives. One way of doing this is by retaining massive amounts of data on consumers. Data collection allows firms to track individual habits on consumption, platform interaction, and much more. As society has grown into this new normal, policymakers have enacted policy to limit the amount of tracking firms can utilize. One dimension along which data protection policy may be important is in advertising and promotion targeting.

To examine the efficacy of these policies, I estimate consumer demand incorporating preferences for variety in consumption. Love of variety seeks to explain why consumers switch products between periods aside from randomness or prices. Primarily researched in the marketing field, implementing a love of variety grants feasibly richer demand estimates, disentangling meaningful consumer preferences from either randomness or other parameters. Much of the prior literature uses brand level product switching to estimate the preference for variety in consumption. Some explanations include attribute fatigue, time varying preferences for attributes, and preferences for quantity in a consumption basket. In this paper, I seek to examine attribute level variety in consumption, allowing me to understand love of variety both within and between products. This preference for variety is the information which the firm uses to target via prices to induce product switching.

Using a model predicated on the idea of fatigue from repeated consumption, I estimate consumer preferences for yogurt flavors with data from Nielsen. I find that love of variety does not meaningfully affect estimates for consumer preferences over product attributes. In fact, I find that the two parameters are difficult to identify under my specification. I also find that consumers experience disutility from continuous consumption, indicating preference for varied purchases to avoid fatigue. When looking at the firm side, I find firms face lower profits under policy which obscures information about consumers. This learning cost can be as large as \$0.02 per unit. The firm loses profit as they lower prices to induce switching and thus learn the distribution of consumer types.

Section two discusses related literature and this paper's contribution, section three examines and explains the data, section four motivates the demand model and informs identification, section five presents the demand model while section six walks through the estimation strategy. Section seven follows with the demand estimation results, section eight introduces the supply model, section nine is comprised of the supply simulation and requisite results. Finally, section ten concludes.

2 Related Literature

The variety-seeking and state dependence literature has been primarily focused on understanding the dynamics of consumer preferences over time. Earlier papers in this literature, like those of McAlister [1982] and Lattin [1987] are most closely linked with this paper's interpretation of variety. They posit that a consumer's preferences for a specific product attribute is satiated with repeated

purchases. McAlister finds her model of dynamic attribute satiation outperforms models with independent choice occasions in accuracy and significance of estimates. Lattin finds that there is heterogeneity in the variety-seeking types, though he uses a tiny cross-section of his own students for the study. The satiation-based approach was then built upon by Givon [1984] who used an inertia model to show how consumers can differ by preference for brand versus preference for variety. By using the brand rather than a specific attribute of a product, Givon is not able to pick up on specific drivers of product switching. Erdem and Keane [1996] also looks at the brand choice problem, modeling decision making when consumers do not have perfect knowledge of preferences over products. The authors use a Bayesian updating framework by which consumers can learn about products via experience or advertising. Their key finding is that consumers are risk averse in consumption, leading them to stay with brands they are familiar with. This model is informative as the reverse-side of my supply-side framework. Marketing researchers, Chintagunta [1999] and Dubé [2004], greatly inform the demand side of this paper. Chintagunta [1999] shows the validity of what he terms the "Lightning Bolt" model of brand choice. He allows for time-varying variety seeking, implying the variety-seeking type can change with time. He finds that state dependence over time and marketing variables are informative in demand estimation. Dubé [2004] progresses Chintagunta's work, showing how consumers seek variety in quantities purchased through the lens of carbonated sodas. Dubé successfully estimates consumer preferences over bundles of products within a category, finding that consumer demographics were key in determining the joint distribution. Both of these papers are primarily concerned with brand choice rather than classifying types of consumers. Finally, Ackerberg [2003] adds to Erdem and Keane's work, allowing for consumers to learn about product attributes via advertising. The key finding here is that switching is primarily induced by the signals put out by advertisements.

The literature on personalized pricing is mostly theoretical for physical goods. Grossman and Shapiro [1984] develops a model of advertising expenditure in which the firm is trying to inform consumers of their products' attributes. Their key finding is that advertising provides a social function by improving consumer and product matching. They also find there are excessive levels of advertising in nearly all cases. Karle and Reisinger model optimal advertising strategies for the firm. Using a theoretical model with two firms, he finds that firms' profits change non-monotonically with targeting ability, decreasing as targeting improves before increasing again. Moorthy and Shahrokhi Tehrani [2023] derives the optimal conditions for targeted advertising. Using a Hotelling framework, the authors find that the key driver of optimal expenditure levels is product differentiation, claiming it as a managerial framework in the face of targeting regulation. Dubé and Misra [2023] uses a randomized control trial (RCT) to understand how price personalization impacts firm profits and consumer welfare. He finds that total consumer surplus declines with personalized pricing, 60% of consumers benefit and, under some conditions, consumer welfare may increase.

My contribution is twofold. Firstly, I am primarily concerned with attribute variety rather than brand variety, where much of the previous work is focused. When looking at variety in attributes, I am able to see how consumers switch both within brand as well as between brands. Further, I

seek to validate love of variety's application in consumer demand. On the supply side, I develop a simple model which can derive optimal pricing depending on consumer switching-type. This model is then used to simulate the effect of data privacy policy around targeted advertising by obscuring the firm's information around consumer types.

3 Data

The data comes from Nielsen's HomeScan and MarketScan. These two main datasets give me access to household purchase decisions as well as product characteristics. From HomeScan, I can access household characteristics, purchase histories by shopping trip, and information on purchase characteristics such as discount use or the total price paid. MarketScan provides product characteristics like price or if the product was in a featured position on the shelf. I limit the sample to Atlanta, Chicago, Denver, Houston, Des Moines, Phoenix, Philadelphia, and San Diego for years 2022-2024. These markets were chosen since they are major cities in different parts of the country. I only include single-person households to avoid any contamination of shopping for multiple people like children or spouses. This ensures I am picking up true preferences for variety rather than buying for multiple people. These trips are encoded with unique identifiers, thus allowing me to look at per-trip, and within-trip, decisions. Agents are under observation for eight weeks and asked to record each shopping trip. Stores are restricted to be valid Nielsen stores, meaning they appear in the retail code data. This is necessary to observe product prices.

I use yogurt data as the product space for my analysis. Yogurt was chosen due to its relative lack of storability, wide menu of flavors, as well as diverse nutrition characteristics. I classify flavors as berry, other flavors, and plain. Plain is a unique identifier while "berry" and "other flavors" are aggregates of many flavors. Berry includes any flavor with "berry" in its flavor encoding, as well as cherry flavors. Other flavors are composed of all flavors which are neither plain nor other. By construction, it is the largest category. The outside option is defined as any purchase which is not yogurt.

Table 1 shows that the full sample of single-person households in the selected markets is 5,408 households. When filtering down to single-serving yogurt purchasing households, this falls to 998 households. These 998 households make more trips overall, earn more, and purchase with coupons less. These households make yogurt purchases in about four weeks on average during their observation period. In a given week, households make about eight yogurt purchases on average. The average income for households that purchase yogurt is between $45,000-49,999 while non-yogurt households is $40,000-44,999. The age of consumers is approximately 55 years old. Households are relatively similar along basic characteristics, especially age, income, and coupon usage.

Figure 1 presents an HHI index for flavors. Over 70% of households are almost entirely loyal to a single flavor. This is likely a byproduct of the simplified aggregation of flavors, allowing "other flavors" to dominate in this measure. Figure 2 further indicates this is occurring. About 60% of consumer baskets are comprised of "other", regardless of purchase volume. Berry flavors are

also relatively stable. Thus, it appears that, on average, plain yogurt is what is substituted for alternative flavor choices. Lastly, figure 3 presents this as the average flavor share by week. The trend is quite similar to what was seen in figure 2, with other flavors dominating, plain being the lowest chosen flavor, and berry being something of a middle ground. The data indicates that other flavors dominates the other two categories and that plain yogurt is the least preferred choice.

4 Demand Model

In each period t, the consumer i chooses a product j from the menu of available products J_t. The available products are $J_t = \{0, 1, 2, 3\}$ where $j = 0$ is the outside option, $j = 1$ is purchasing berry yogurt, $j = 2$ is purchasing plain yogurt, and $j = 3$ is purchasing other flavors. Consumers are assumed to be state-dependent, carrying previous choices forward when love of variety is present. Otherwise, they only internalize characteristic utility and price utility. Each product $j \in J_t$ is a bundle of $k \in K$ characteristics such that $X_{jt}^k \in \mathcal{X}_{jt}$. The consumer's utility from choosing product j at time t is given by:

$$U_{ijt} = \beta_0 + \beta_{it} X_{jt} + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \varepsilon_{ijt}, \text{ for j} \in \{1, 2, 3\}$$

such that

$$\Xi_{ijt} = \lambda \sum_{s=1}^{8} \delta^{t-s} \Xi_{ijs} + (1 - \lambda) \sum_{j \neq k} \rho x_{k,t-1}$$

$$\varepsilon_{ijt} \sim \text{T1EV}(0, 1)$$

$$U_{i0t} = \varepsilon_0$$

In the utility function, there are the following components: β_0 is a constant, β_{it} is a consumer's utility for characteristic X_{jt} associated with product j in time t. Following Lattin [1987], love of variety preference is captured by γ_i. A consumer's history is given by Ξ_{ijt} where λ is a weight on repeated consumption Ξ_{ijs} which is discounted with time δ^{t-s}. ρ is an indicator for if a consumer bought a product $x_{k,t-1}$ where $k \neq j$. α_i is the parameter on prices p_{jt} and ε_{ijt} is the Gumbel shock.

Choice probabilities take the following form:

$$P_{ijt} = \frac{\exp(\beta_0 + \beta_{it} X_{jt} + \gamma_i \Xi_{ijt} + \alpha_i p_{jt} + \varepsilon_{ijt})}{1 + \sum_k \exp(\beta_0 + \beta_{it} X_{kt} + \gamma_i \Xi_{ikt} + \alpha_i p_{kt} + \varepsilon_{ikt})}$$

Where we normalize the outside option, not purchasing yogurt, to be zero by construction.

5 Motivation and Identification

Before estimating the model, I run a Monte Carlo simulation to see how the model responds to the addition of love of variety. This controlled setting allows me to verify that my parameters, β, γ can be separately identified. The simulation also reveals if there is omitted variable bias in the specification without love of variety.

The simulation sets the following parameters:

Simulation Parameters

Parameter	**Value**	**Description**
Panel & Simulation Dimensions		
S	1000	Number of Monte Carlo simulation runs
T	100	Total evaluation time periods per run
T_{prior}	5	History formation / burn-in period
J	5	Number of available products in choice set
Structural Preference & Satiation Parameters		
$\mathcal{X}$	$[1.0, 5.0]$	Linear product space
β	2.0	Baseline quality utility parameter
γ	$\{0, 3, 6, 9\}$	Variety utility / satiation penalty grid
ϵ_{ijt}	Gumbel$(0, 1)$	Unobserved i.i.d. extreme value preference shock

From table 2, I find that choice probabilities are biased when love of variety is not accounted for. When $\gamma = 0$, 75% of the probability mass is on product 5. This makes sense, product 5 has the highest "value", thus without preferences for variety, a majority of consumers would choose it. However, as the magnitude of love of variety increases, we see the probability distribution shift. At a low level of variety, $\gamma = 3$, product 5 still dominates, but products 1 and 2 now see massive increases in probability. This is due to variety in this setting meaning absolute distance and thus pushing consumers towards the extremes of the product distribution. This trend is exacerbated as the magnitude of γ increases. When $\gamma = 12$, the highest value allowed, the probability is focused on the poles of the distribution, with a nearly 40/60 split between products 1 and 5, respectively.

Figure 4 shows this trend graphically. Note that after about 10-15 periods, consumers reach a steady state in consumption when there is love of variety. In the specification without love of variety, the choice probabilities are in rank order by utility and do not change much from the beginning of time. As we increase the level of γ, we see product 5's choice probability reduced from approximately 80% to the near 60% discussed above. The graph also shows that γ determines the menu of products actually under consideration to a consumer. For a consumer with low love of variety, they will consume all at differing levels. For medium love of variety, consumers only purchase products 1, 2, and 5. At high levels, this goes to only products 1 and 5. Thus, their preferences become more particular.

This exercise points to love of variety being meaningful to include in demand estimation. When we omit love of variety, we bias preferences for goods in both directions. From this example, good

1 is biased down while goods 3, 4, and 5 are biased upwards. Good 2 is an interesting case, with the direction of bias depending on the level of love of variety. Thus, we have evidence of omitted variable bias in the absence of love of variety, informing the feasibility of estimating love of variety in yogurt consumption with Nielsen data.

6 Estimation

Following Petrin and Train [2010], I use a control function to control for price endogeneity. I do a first stage instrumental variable estimation using Hausman instruments and then take residual from that estimation and insert it into the utility function. The first stage takes the form:

$$p_{jt} = \bar{p}_{j,-m(i),t} + \tau_t + \eta_{jt}$$

Where $\bar{p}_{j,-m(i),t}$ is the mean price from the three markets the household is not shopping in and τ_t is a time indicator. The mean price of the other three markets was chosen as an instrument to separate any region specific effects like local advertising. Thus, I assume there is no national advertising or other effects which impact demand across all markets. The residual recovered from the first stage corrects for the bias present in pricing, allowing us a consistent estimate of pricing.

I make an index of prices, instrumental variable residuals, and previous choices for each household's trips for each flavor. These are then used to estimate the objective function:

$$U^k_{ijt} = \beta_0 + \sum_k \beta^k_i X^k_{jt} + \gamma_i \Xi_{ijt} + \alpha p_{jt} + \sigma \hat{\eta}_{jt}$$

Where β_0 is a constant, β^k for k $\in$ {berry, other, plain} are choice specific coefficients for choosing berry, other, or plain. γ is the love of variety term, interacted with Ξ_{ijt}, which captures if consumer i switched between periods. The price and control function term are unchanged from above. The probability of individual i choosing the observed product j in time t is given by:

$$\Pi_i (P_{ijt}(\beta_i t, \Xi_{ijt}))^{y_{ijt}}$$

Where $y_{ijt} = 1$ when person i chosen j that period, and zero otherwise. The log-likelihood is then given by:

$$\text{LL}(\theta) = \sum_t \sum_i \sum_j y_{ijt} \ln P_{ijt}(\beta_i, \Xi_{ijt})$$

I set $\lambda = 0.7$, following previous literature, and $\delta = 0.9$ for the same reason. I repeat estimation under specifications which exclude γ_i, do not estimate heterogeneous β_i, γ_i, and do both. This allows me to see if results are biased when love of variety is excluded from estimation.

7 Demand Results

Table 4 shows the main results from the estimation. All coefficients are flavor-specific and are relative to the "other flavors" category which is relative to the outside option. When looking at the standard logit, without heterogeneous consumers, I find that consumers experience disutility from both berry flavored and plain yogurt compared to other flavors. They also experience disutility from continuing to consume the same flavor, with berry inducing more disutilty than plain, though plain is not statistically significant. Price effects are negative and significant, with the control function residual being positive and significant.

From the mixed logit, I find similar estimates. I recover statistically significant coefficients for all but the coefficient on variety for plain yogurt and the estimated standard deviation on γ for berry. When looking at the coefficient on berry, I find a large and significant disutility for continuing to consume berry yogurt given past berry utility. I also find a smaller disutility for continuing to consume plain yogurt. The plain fatigue coefficient is not significant, pointing to identification issues. The estimated standard deviations for the beta's imply that there is little heterogeneity in preferences for flavors once love of variety is introduced. Prices are still significant and negative, though less than in the standard logit specification along with the control function residual being significant and more positive. I also find that implied willingness to pay does not change much for flavor preferences between the standard and mixed logit specification. However, plain yogurt consumers are willing to pay a lot to switch from their flavor, needing over \$7 to stay with plain. This points to relatively large discounts on plain yogurt being required to overcome the boredom in flavor. This is massive compared to the, insignificant, estimate in the standard case.

Looking at table 3, I find the estimates and implied willingness to pay for what I term the "naive" specifications. Here, there is no love of variety γ. When comparing the estimates across specifications, I see little change in the magnitudes of the estimates. For plain, I find a 0.01% change in base utility for consuming plain yogurt in the standard logit and 0.19% in the mixed logit. For berry, the decrease is 0.78% in the standard logit and a 1.87% decrease in the mixed logit. The intercept has a difference of 0.2% in the standard logit and 0.47% in the mixed while price is an increase of 0.15% and 0.33%, respectively.

These results point to love of variety not being meaningfully additive to demand estimation, at least in the specification I have opted for. For one, there are clear signs that the γ terms are not identified, having either massive standard errors on the coefficient itself or large estimated standard deviations.

8 Supply Side

There are N consumers, each endowed with a type γ_r for $r \in \{H, L\}$, corresponding to high or low utility from variety in products consumed. As such, there are $N_r \in N$ consumers of each type, $r \in \{H, L\}$. Consumers maximize utility according to the following:

$$U_{irjt} = \beta_0 + \gamma_r\{j_t \neq j_{t-1}\} + \alpha p_{rjt} + \varepsilon_{irjt}$$

Where utility is over individuals $i \in N$, type $r \in \{H, L\}$, products $j \in J$, and time $t \in T$. β_0 is an intercept, representing the baseline utility from consuming. γ_i is the consumer's utility from consuming different products, and is active if they switched products from one period to the next. α is disutility of price, which can be lowered via a discount from advertising. Finally, ε_{ijt} is the Gumbel error, giving us a closed form Logit.

This equation can be written in terms of deterministic vs. random utility:

$$U_{irjt} = V_{irjt} + \varepsilon_{irjt}$$
$$\text{s.t.}$$
$$V_{irjt} = \beta_0 + \gamma_r\{j_t \neq j_{t-1}\} + \alpha(p_{jt} - d_{rjt})$$

Choice probabilities of consuming the inside option(s), as well as the outside option utility normalized to be zero, is written as such:

$$P_{irjt} = \frac{e^{V_{irjt}}}{1 + \sum_k e^{V_{irkt}}}$$
$$P_{ir0t} = \frac{1}{1 + \sum_k e^{V_{irkt}}}$$

Next, because consumers of the same type r are homogenous:

$$s_{rjt} = \sum_i P_{irjt} \text{ for r} \in \{\text{H,L}\}$$

Taking the derivative with respect to prices, we can get the own and cross price elasticities:

$$\epsilon^p_{jj} = \alpha p_{jt}(1 - s_{rjt})$$
$$\epsilon^p_{jk} = -\alpha p_{kt} s_{rkt} \quad \text{for k}\neq\text{j}$$

Thus, since $\alpha < 0$, we can see that own price elasticity is negative and cross-price is positive.

Now, a supermarket enters the market. The supermarket is able to control prices $\bar{p}_j$ for a product j. The supermarket can also offer a discount d_{rjt} which is tailored to our switching types $r \in \{H, L\}$. The firm's actual price is written as:

$$p_{rjt} \equiv \bar{p}_j - d_{rjt}$$
$$\text{s.t.}$$
$$d_{rjt} > 0$$

So, as the market raises the discount, utility increases as $\alpha < 0$ so $-\alpha d_{rjt} > 0$. It is through this discounting that firm can induce switching. The firm's profit margin for a product j is given by

$$m_{rjt} = p_{rjt} - c_j$$
$$= \bar{p}_j - d_{rjt} - c_j$$

This says that as we raise the discount, we raise market share on the product being discounted but lose profit margin.

With these definitions in mind, we will be looking at a two period model under two specifications. First, the case where the firm has perfect information on the consumers' types and can thus perfectly price. In the second case, we will look at the case of imperfect information, where the firm is using Bayesian updating to approximate the type distribution.

8.1 Perfect Information

Assume the firm has perfect information over the distribution of types. That is, the supermarket perfectly observes each consumer type r and thus knows the mass N_r of each type. Define period one profit, π_1 as such:

$$\pi_1 = \sum_r N_r \sum_j s_{rj1}(m_{rj1})$$
$$\text{s.t.}$$
$$s_{rj1} = P_{irj1}$$
$$m_{rj1} = \bar{p}_j - d_{rj1} - c_j$$

Here, the firm is understanding how each type's share for each product impacts their profits. We can similarly define period two profits, π_2 like so:

$$\pi_2 = \sum_r N_r \sum_j s_{rj1}[\sum_k s_{rk2}(j)(m_{rk2}(j))]$$

The only augmentation is the firm now understands all of the possible paths for a consumer of

type r who chose product j in period 1, and product k in period 2. For the sake of conciseness in notation, $w_r(j) = \sum_k s_{rk2}(j)(m_{rk2}(j))$. Define total profit, Π, as such:

$$\begin{aligned}\Pi &= \pi_1 + \pi_2 \\ &= \sum_r N_r \sum_j s_{rj1}(m_{rj1} + \delta w_r(j))\end{aligned}$$

Where δ is a discount factor on future profits. Taking the derivative with respect to period one prices for a product j yields:

$$\frac{\partial \Pi}{\partial p_{rj1}} : 0 = 1 + \alpha \sum_{k \neq j} s_{rk1}[(m_{rj1} - m_{rk1}) + \delta(w_r(j) - w_r(k))]$$

The firm sets prices according to the current difference in profit margins and the difference in future profits which occur upon switching a consumer between product j and product k. When we do the same for period two, we recover the following:

$$\frac{\partial \Pi}{\partial p_{rk2}} : 0 = 1 + \alpha \sum_{l \neq k} s_{rl2}(j)[(m_{rk2}(j) - m_{rl2}(j))]$$

Now, expanding the expression and solving for period one prices:

$$\begin{aligned}
0 &= 1 + \alpha[\sum_{k \neq j} s_{rk1}((p_{rj1} - c_j) - (p_{rk1} - c_k) + \delta(w_r(j) - w_r(k)))] \\
0 &= 1 + \alpha(p_{rj1} - c_j)\sum_{k \neq j} s_{rk1} - \alpha \sum_{k \neq j} s_{rk1}(p_{rk1} - c_k) + \alpha\delta \sum_{k \neq j} s_{rk1}(w_r(j) - w_r(k)) \\
\alpha(p_{rj1} - c_j)\sum_{k \neq j} s_{rk1} &= -1 + \alpha \sum_{k \neq j} s_{rk1}(p_{rk1} - c_k) - \alpha\delta \sum_{k \neq j} s_{rk1}(w_r(j) - w_r(k)) \\
\alpha p_{rj1}\sum_{k \neq j} s_{rk1} &= \alpha c_j \sum_{k \neq j} s_{rk1} - 1 + \alpha \sum_{k \neq j} s_{rk1}(p_{rk1} - c_k) - \alpha\delta \sum_{k \neq j} s_{rk1}(w_r(j) - w_r(k)) \\
p^*_{rj1} &= c_j - \frac{1}{\alpha \sum_{k \neq j} s_{rk1}} + \frac{\sum_{k \neq j} s_{rk1}(p_{rk1} - c_k)}{\sum_{k \neq j} s_{rk1}} - \delta \frac{\sum_{k \neq j} s_{rk1}(w_r(j) - w_r(k))}{\sum_{k \neq j} s_{rk1}} \\
p^*_{rj1} &= c_j - \frac{1}{\alpha(1 - s_{rj1})} + \frac{\sum_{k \neq j} s_{rk1} m_{rk1}}{1 - s_{rj1}} - \delta \frac{\sum_{k \neq j} s_{rk1}(w_r(j) - w_r(k))}{1 - s_{rj1}}
\end{aligned}$$

From this, the firm with perfect information sets each type r's price to equal the cost plus a demand markup and the share weighted average margin of alternatives $k \neq j$ minus the intertemporal tradeoff. Here, the firm will be willing to lower the price of good j today if it implies higher profits

tomorrow ($w_r(j) > w_r(k)$), increasing probability of purchase. In the opposite case, the firm would raise prices on product j to encourage shifting from product j to product k tomorrow. That is, they use the past choice to determine consumer state and thus future profits for each product. The firm again thinks of the tradeoff, without having to think about future profit as period two is terminal. Under perfect information, the supermarket chooses its prices by evaluating the current margin and future profit associated with reallocating consumers across products. Now, we move to the case of imperfect information.

8.2 Imperfect Information

In this case, the supermarket does not know consumer types with certainty, nor do they know the mass of either type. So, they must infer the distribution from from consumer purchases. Define the following objects:

$$\lambda_1 = Pr(r = H)$$
$$\lambda_2 = Pr(r = H|j) \Rightarrow \lambda_2(j) = \frac{\lambda_1 s_{Hj1}}{\lambda_1 s_{Hj1} + (1 - \lambda_1) s_{Lj1}}$$
$$\tilde{s}_{j1} = \lambda_1 s_{Hj1} + (1 - \lambda_1) s_{Lj1}$$
$$\tilde{s}_{k2}(j) = \lambda_2(j) s_{Hk2}(j) + (1 - \lambda_2(j)) s_{Lk2}(j)$$
$$\mathbb{A} = [\lambda_1 s_{Hj1}(1 - s_{Hj1}) + (1 - \lambda_1) s_{Lj1}(1 - s_{Lj1})]$$

Now, we can define the profit functions much like we did before:

$$\pi_1 = N \sum_j \tilde{s}_{j1} m_{j1}$$
$$\pi_2 = N \sum_j \tilde{s}_{j1} \sum_k \tilde{s}_{k2}(j) m_{k2}(j)$$

The key difference here is that the firm must now update their beliefs from period to period rather than being able to discount according to the tradeoffs between two products. This implies that there is room for error in discounting. When combining the two into overall profit, we get:

$$\Pi = N \sum_j \tilde{s}_{j1} [m_{j1} + \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j)]$$

Now, we can take the derivative with respect to period one prices for product j:

$$\frac{\partial \Pi}{\partial p_{j1}} : 0 = N\tilde{s}_{j1} + \alpha N[\lambda_1 s_{Hj1}(1 - s_{Hj1}) + (1 - \lambda_1)s_{Lj1}(1 - s_{Lj1})][m_{j1} + \delta \sum_k \tilde{s}_{k2}(j)m_{k2}(j)]$$
$$- \alpha N \sum_{h \neq j}[\lambda_1 s_{Hh1}s_{Hj1} + (1 - \lambda_1)s_{Lh1}s_{Lj1}][m_{h1} + \delta \sum_k \tilde{s}_{k2}(h)m_{k2}(h)]$$
$$+ \delta \sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}}[\pi_{H2}(h) - \pi_{L2}(h)]$$

This is much the same as the perfect information case. However, looking at the third line, we can see where the firm incorporates their belief updating. Define the derivative in the third line as such:

$$\frac{\partial \lambda_2(h)}{\partial p_{j1}} = \begin{cases} 0 = \frac{\alpha\lambda_1(1-\lambda_1)}{\tilde{s}_{j1}^2}[s_{Hj1}s_{Lj1}(s_{Lj1} - s_{Hj1})] \text{ if } h = j \\ 0 = \frac{\alpha\lambda_1(1-\lambda_1)}{\tilde{s}_{h1}^2}[s_{Hh1}s_{Lh1}(s_{Lj1} - s_{Hj1})] \text{ if } h \neq j \end{cases}$$

From this, the firm updates its belief about the distribution of consumer types upon observing period one consumption. The magnitude and direction of this update depends on how likely that choice is for each type. In total, the third line of the derivative interacts the probability of purchasing item h with the effect of price on the firm's posterior belief and the value of its belief in one type or the other.

Now, we can solve for p_{j1} via the following steps. First,

$$\alpha N\mathbb{A}[p_{j1} - c_j + \delta \sum_k \tilde{s}_{k2}(j)m_{k2}(j)] =$$
$$- N\tilde{s}_{j1} + \alpha N \sum_{h \neq j}[\lambda_1 s_{Hh1}s_{Hj1} + (1 - \lambda_1)s_{Lh1}s_{Lj1}][m_{h1} + \delta \sum_k \tilde{s}_{k2}(h)m_{k2}(h)]$$
$$- N\delta \sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}}[\pi_{H2}(h) - \pi_{L2}(h)]$$

Now, we isolate p_{j1}:

$$\alpha N\mathbb{A}p_{j1} = \alpha N\mathbb{A}c_j - \alpha N\mathbb{A}\delta \sum_k \tilde{s}_{k2}(j)m_{k2}(j)$$
$$- N\tilde{s}_{j1} + \alpha N \sum_{h \neq j}[\lambda_1 s_{Hh1}s_{Hj1} + (1 - \lambda_1)s_{Lh1}s_{Lj1}][m_{h1} + \delta \sum_k \tilde{s}_{k2}(h)m_{k2}(h)]$$
$$- N\delta \sum_h \tilde{s}_{h1}[\frac{\partial \lambda_2(h)}{\partial p_{j1}}(\pi_{2H}(h) - \pi_{2L}(h))]$$

And simplify:

$$
\begin{aligned}
p_{j1} = c_j &- \delta \sum_k \tilde{s}_{k2}(j) m_{k2}(j) - \frac{\tilde{s}_{j1}}{\alpha A} \\
&+ \frac{\sum_{h \neq j} [\lambda_1 s_{Hh1} s_{Hj1} + (1-\lambda_1) s_{Lh1} s_{Lj1}]}{\mathbb{A}} [m_{h1} + \delta \sum_k \tilde{s}_{k2}(h) m_{k2}(h)] \\
&- \frac{\delta}{\alpha \mathbb{A}} [\sum_h \tilde{s}_{h1} \frac{\partial \lambda_2(h)}{\partial p_{j1}} (\pi_{2H}(h) - \pi_{2L}(h))]
\end{aligned}
$$

Deconstructing the optimal price, optimal price is composed of the marginal cost, minus the own-product discounted continuation value, plus a demand markup term and the the value of alternative products, and lastly a term referring to the learning adjustment from period one behavior. In the imperfect-information specification, today's prices affect current demand and the distribution of future states as well as the information that can be learned from consumer purchases. Period one prices alter the likelihood of purchase, thus affecting the posterior over consumer type and, by extension, optimal pricing in period two.

9 Supply Simulation

In the supply simulation, I set the following parameters

Model Parameters

Parameter	**Symbol**	**Assigned Value**
Base Brand Utilities	β_0	$[2.2, 2.0]$
High-Type Switching Utility	γ_H	2.0
Low-Type Switching Utility	γ_L	0.1
Price Sensitivity Coefficient	α	-1.5
Discount Factor	δ	0.95
Marginal Unit Costs	c	$[1.2, 1.0]$
Number of Inside Goods	J	2
Initial State Distribution	s_0	$[0.5, 0.5]$
Price Optimization Bounds	$[P_{\min}, P_{\max}]$	$[\$1.00, \$8.00]$

Then, I give consumers a prior j_0 so period one prices are informed by switching in the imperfect case. In table 5, the optimal prices, posterior beliefs, and profits are displayed. Firms lower prices on the lower cost good, directing consumers to it to raise profits. Firms offer higher discounts when the distribution of consumers has less high-switchers, needing to induce low-types to switch via price effects. The firm consistently overestimates the amount of high types in the population by a magnitude of anywhere from 10-15 percent. We see profits maximized in the case akin to perfect information where all individuals are high type. When comparing the case of perfect and imperfect information profits, the value of information is between 1.2-1.9 cents per unit. This cost of information is maximized when the distribution is evenly split, implying the learning adjustment cost from the imperfect cost specification is meaningfully impacting firm profits.

This result points to consumer privacy policies being a viable way to limit tracking a consumer's data. Firms who are unwilling to incur the losses of learning will stop tracking as diligently and the firms large enough to weather the losses will need some other form of enforcement (fines, bans, etc.). In physical goods spaces, this shift to privacy is perhaps less poignant. In digital markets, firms are able to track every action a consumer takes on their domain and can use that information to target advertising, discounts, or otherwise induce prolonged usage of the product.

10 Conclusion

I implement a dynamic discrete choice model of consumer demand which incorporates love of variety in consumers. The model is estimated with Nielsen's HomeScan and MarketScan data, allowing for panel analysis of about 1,000 single agent households. I find that love of variety does not meaningfully impact demand estimates for preferences over product attributes, with the largest effect being a 1.9% decrease in the magnitude of the estimate for baseline preference of berry yogurt. My findings also show that consumers need to be compensated to continue to consume flavors they have previously consumed. This indicates love of variety is actually fatigue avoidance. However, I put forth that love of variety is not properly identified in my current specification, implying further tuning is needed.

On the supply side, I derive optimal pricing on the part of the firm using a simple toy model. Under a regime in which the firm has perfect information on the distribution of types, they optimally price discriminate, pushing consumers to the lower-cost good. However, when policy which obscures information is introduced, I find that firms lose up to \$0.02 per unit due to the cost of learning consumer types. As data privacy policy continues to be enacted, this finding points to its efficacy in protecting consumers.

In closing, this paper has contributed to the love of variety and targeted advertising literatures. Further work can be done to tune the demand model for more precise estimation, add characteristics over which consumers can have preferences, and better control for price endogeneity. On the supply side, calibrating the model with MarketScan data and consumer purchases observed in HomeScan would allow for richer findings from the optimal pricing model. Joining the supply and demand sides

into a partial equilibrium model would be of great interest and provide opportunity for counterfactual analyses.

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11 Tables and Figures

Table 1: Summary Statistics: Full Sample vs. Estimation Sample

Statistic / Metric	Full Panel Sample	Estimation Sample
Unique Households	5,408	998
Mean Total Store Trips / HH	308.15	400.05
Mean Observed Choice Weeks / HH	N/A	3.91
Mean Household Income	$40,000-44,999	$45,000-49,999
Mean Head Age	50-55 yrs	50-55 yrs
Deal / Coupon Purchase Rate	26.52%	23.10%
Mean Weekly Yogurt Units (Purchasers)	N/A	8.44 units

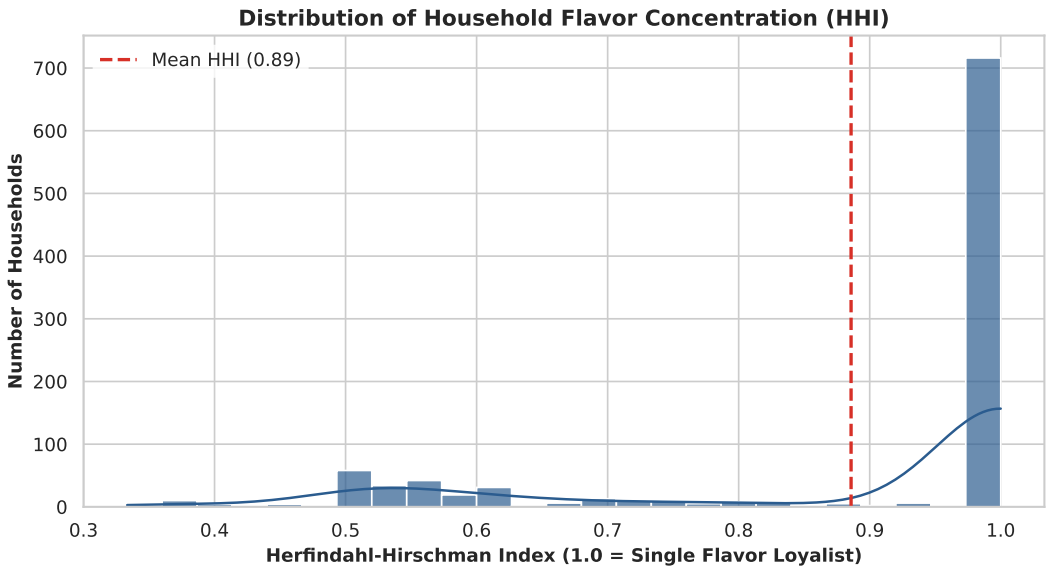


Figure 1: Distribution of Household Flavor Concentration (HHI)

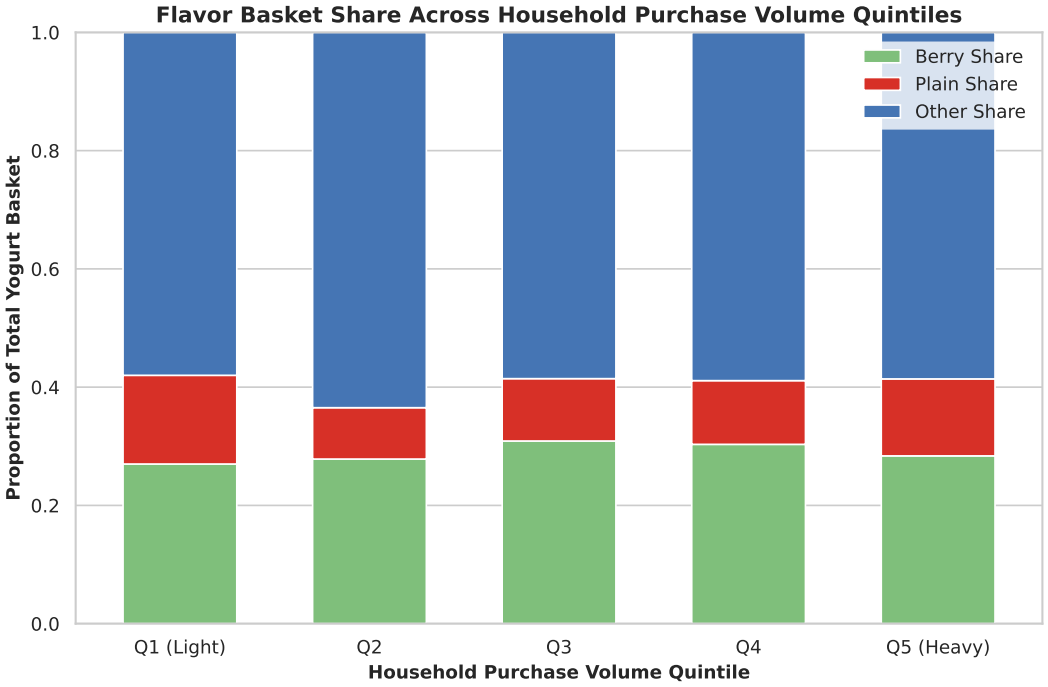


Figure 2: Flavor Basket Share Across Household Purchase Volume Quintiles

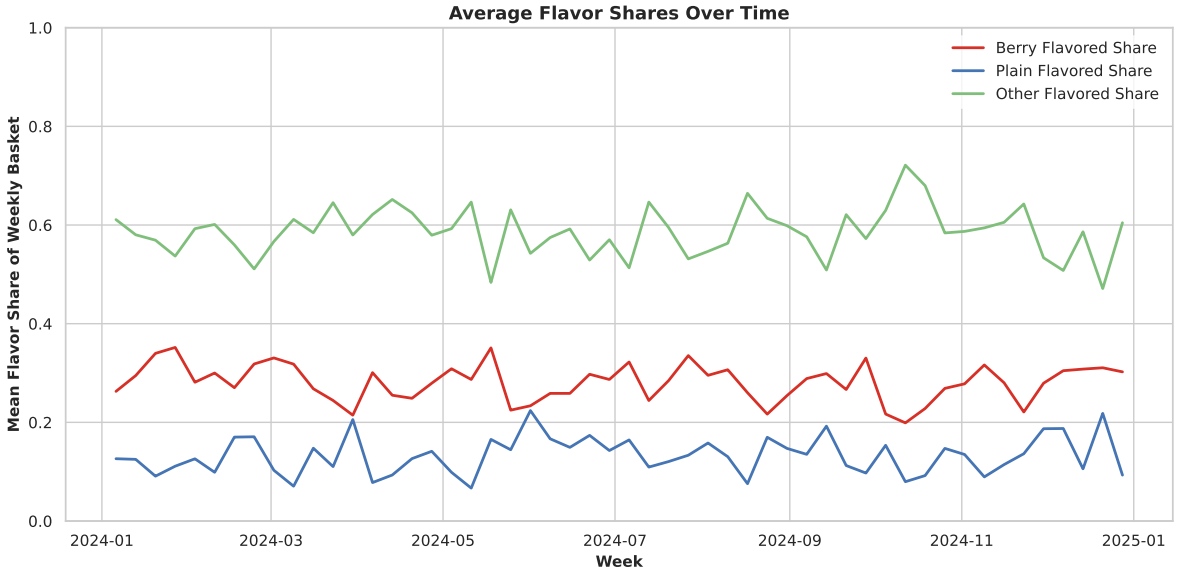


Figure 3: Average Flavor Shares Over Time

Table 2: CCP Comparison Across Levels of Variety Utility

	Variety Utility (γ)			
Metric / Choice Option	$\gamma = 0$	$\gamma = 3$	$\gamma = 6$	$\gamma = 9$
Choice Probabilities (CCPs)				
Product 1 ($X = 1.0$)	0.0016	0.1352	0.3321	0.3973
Product 2 ($X = 2.0$)	0.0084	0.1325	0.0541	0.0084
Product 3 ($X = 3.0$)	0.0437	0.0844	0.0026	0.0000
Product 4 ($X = 4.0$)	0.1934	0.0553	0.0023	0.0002
Product 5 ($X = 5.0$)	0.7529	0.5926	0.6089	0.5941
Welfare / Inclusive Value				
Inclusive Value (Mean)	10.9606	13.0592	17.4586	22.4415

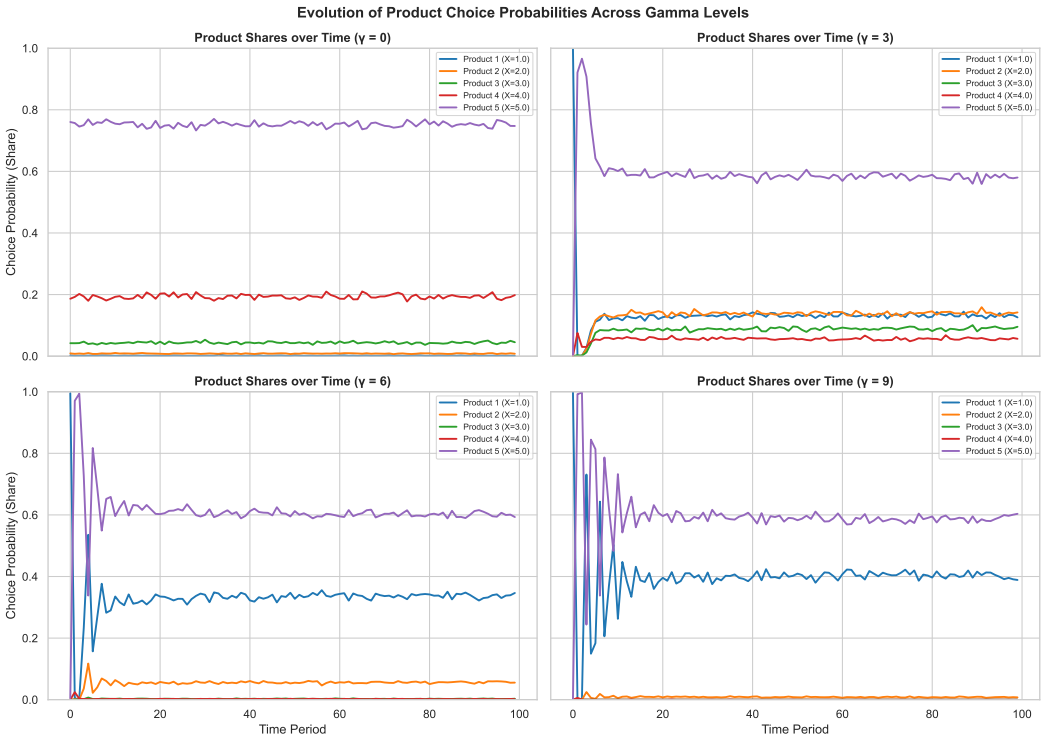


Figure 4: Choice Probabilities Over Time

Table 3: Naive Demand Model Estimates (No Variety)

	Standard Logit			Random Coefficient Logit		
Parameter	**Estimate**	**Std. Error**	***p*-value**	**Estimate**	**Std. Error**	***p*-value**
Constant (α_0)	0.9455	0.0397	$< 0.001^{***}$	0.9459	0.0397	$< 0.001^{***}$
Mean β_{berry}	-0.7635	0.0126	$< 0.001^{***}$	-0.7632	0.0127	$< 0.001^{***}$
Mean β_{plain}	-1.3899	0.0179	$< 0.001^{***}$	-1.3930	0.0218	$< 0.001^{***}$
SD β_{berry}	–	–	–	0.0008	0.0304	0.9785
SD β_{plain}	–	–	–	0.0303	0.1041	0.7711
Price (α)	-0.9836	0.0299	$< 0.001^{***}$	-0.9836	0.0299	$< 0.001^{***}$
Control Function (σ)	0.1670	0.0292	$< 0.001^{***}$	0.1654	0.0292	$< 0.001^{***}$
Log-Likelihood Objective		71955.73			71955.75	
Optimization Success		True			True	
Willingness to Pay (WTP in $)						
WTP: Berry Preference		$0.78			$0.78	
WTP: Plain Preference		$1.41			$1.42	

$^{***}p < 0.001$, $^{**}p < 0.05$, $^{*}p < 0.10$

Table 4: Flavor Demand Demand Model Estimates

	Standard Logit			Random Coefficient Logit		
Parameter	**Estimate**	**Std. Error**	***p*-value**	**Estimate**	**Std. Error**	***p*-value**
Constant (α_0)	0.9436	0.0396	$< 0.001^{***}$	0.9415	0.0396	$< 0.001^{***}$
Mean β_{berry}	-0.7576	0.0127	$< 0.001^{***}$	-0.7489	0.0128	$< 0.001^{***}$
Mean β_{plain}	-1.3897	0.0180	$< 0.001^{***}$	-1.3956	0.0264	$< 0.001^{***}$
Mean γ_{berry}	-0.7262	0.1485	$< 0.001^{***}$	-7.2030	1.5424	$< 0.001^{***}$
Mean γ_{plain}	-0.0351	0.1009	0.7278	-0.0555	0.1032	0.5909
SD β_{berry}	–	–	–	1.0513	0.0916	$< 0.001^{***}$
SD β_{plain}	–	–	–	1.0513	0.1492	$< 0.001^{***}$
SD γ_{berry}	–	–	–	148.4132	155.3868	0.3395
SD γ_{plain}	–	–	–	1.0513	0.1641	$< 0.001^{***}$
Price (α)	-0.9821	0.0299	$< 0.001^{***}$	-0.9804	0.0299	$< 0.001^{***}$
Control Function (σ)	0.1667	0.0292	$< 0.001^{***}$	0.1662	0.0292	$< 0.001^{***}$
Log-Likelihood Objective		71934.90			71887.59	
Optimization Success		True			True	
Willingness to Pay (WTP in $)						
WTP: Berry Preference		$0.77			$0.76	
WTP: Plain Preference		$1.42			$1.42	
WTP: Berry Satiation Disutility		$0.74			$7.35	
WTP: Plain Satiation Disutility		$0.04			$0.06	

$^{***}p < 0.001$, $^{**}p < 0.05$, $^{*}p < 0.10$

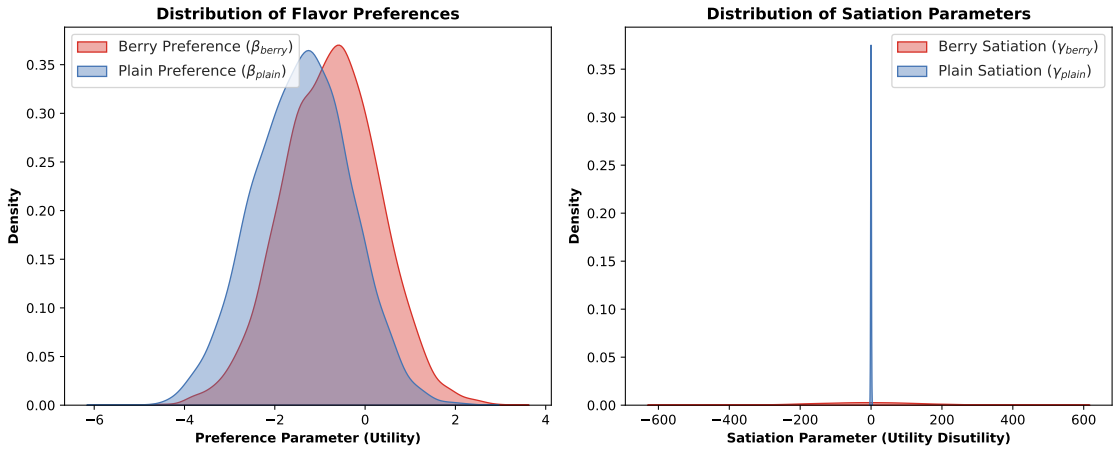


Figure 5: Marginal Type Distributions (β and γ Parameters)

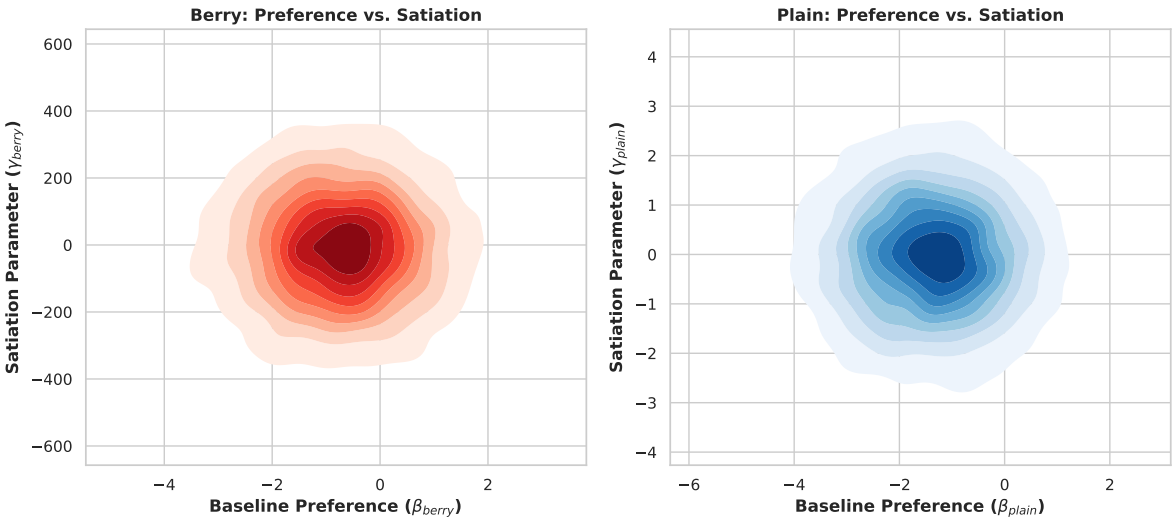


Figure 6: Joint Type Distributions (Preference β vs. Satiation γ Contours)

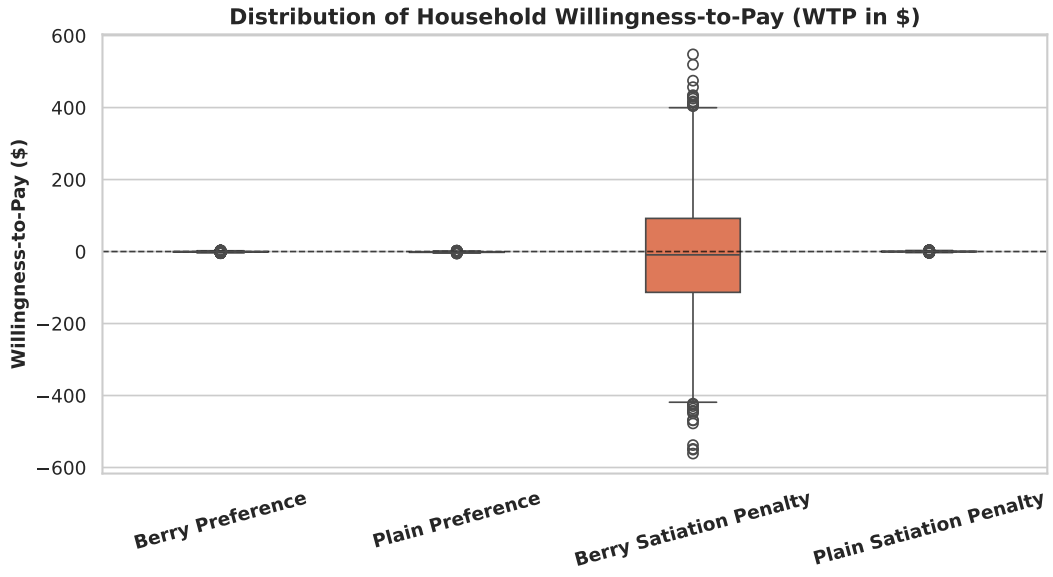


Figure 7: Household Willingness-to-Pay Distributions (WTP in $)

Table 5: Information Structure, Optimal Pricing, and Value of Information Across Population Mixes

Population Mix	**% High** $\left(\frac{N_H}{N}\right)$	**% Low** $\left(\frac{N_L}{N}\right)$	**Imperfect Price (P_1)** **Good 1**	**Good 2**	**Posterior Beliefs** **G1**	**G2**	**Imperfect Profit**	**Perfect Profit**	**Value of Info**
100% LOW	0%	100%	2.0633	1.8646	0.0000	0.0000	0.6031	0.6031	-0.0000
25% HIGH, 75% LOW	25%	75%	2.1847	1.9944	0.3727	0.3641	0.7820	0.7999	0.0178
50% HIGH, 50% LOW	50%	50%	2.2819	2.1049	0.6510	0.6448	0.9773	0.9966	0.0194
75% HIGH, 25% LOW	75%	25%	2.3542	2.1887	0.8528	0.8504	1.1811	1.1934	0.0123
100% HIGH	100%	0%	2.4067	2.2494	1.0000	1.0000	1.3902	1.3902	0.0000